

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT
on

OPERATING SYSTEMS

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

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B. M. S. College of Engineering,
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Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by Yogesh Kumar Sheoran (1BF24CS337), who is Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026

. The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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1.	Write a C program to simulate the following non-pre-emptive CPU scheduling algorithm to find turnaround time and waiting time. →FCFS → SJF (pre-emptive & Non-preemptive) → Priority (pre-emptive & Non-pre-emptive) →Round Robin (Experiment with different quantum sizes for RR algorithm)	1-21
2.	Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.	21-26
3.	Write a C program to simulate Real-Time CPU Scheduling algorithms: a) Rate- Monotonic b) Earliest-deadline First c)Proportional scheduling	26-32
4.	Write a C program to simulate a)producer-consumer problem using semaphores b) Dining Philosophers problem.	32-38
5.	Write a C program to simulate: a) Bankers’ algorithm for the purpose of deadlock avoidance b) Deadlock Detection	39-46
6.	Write a C program to simulate the following contiguous memory allocation techniques. a) Worst-fit b) Best-fit c) First-fit	46-52
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Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating System.
C04	Conduct practical experiments to implement the functionalities of Operating system.

Program -1

Question:

Write a C program to simulate the following non-pre-emptive CPU scheduling algorithm to find turnaround time and waiting time.

→FCFS

→ SJF (pre-emptive & Non-preemptive)

Code:

=>FCFS:

```
#include <stdio.h>

void sort(int pid[], int at[], int bt[], int n)
{
    int i, j, temp;

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(at[j] < at[i])
            {
                temp = at[i];
                at[i] = at[j];
                at[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;

                temp = pid[i];
                pid[i] = pid[j];
                pid[j] = temp;
            }
        }
    }
}

int main()
{
    int n, i;

    printf("Enter number of processes: ");
    scanf("%d", &n);
```

```

int pid[n], at[n], bt[n], ct[n], tat[n], wt[n];

for(i = 0; i < n; i++)
    pid[i] = i + 1;

printf("Enter arrival times:\n");
for(i = 0; i < n; i++)
    scanf("%d", &at[i]);

printf("Enter burst times:\n");
for(i = 0; i < n; i++)
    scanf("%d", &bt[i]);

sort(pid, at, bt, n);

if(at[0] > 0)
    ct[0] = at[0] + bt[0];
else
    ct[0] = bt[0];

for(i = 1; i < n; i++)
{
    if(ct[i - 1] >= at[i])
        ct[i] = ct[i - 1] + bt[i];
    else
        ct[i] = at[i] + bt[i];
}

for(i = 0; i < n; i++)
{
    tat[i] = ct[i] - at[i];
    wt[i] = tat[i] - bt[i];
}

double avg_tat = 0, avg_wt = 0;

printf("\nFCFS Scheduling\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

for(i = 0; i < n; i++)
{
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i], at[i], bt[i], ct[i], tat[i], wt[i]);

    avg_tat += tat[i];
    avg_wt += wt[i];
}

avg_tat /= n;
avg_wt /= n;

printf("\nAverage Turnaround Time = %.2lf ms\n", avg_tat);
printf("Average Waiting Time    = %.2lf ms\n", avg_wt);

return 0;
}

```

Result:

```
Enter number of processes: 4
Enter arrival times:
0 1 2 3
Enter burst times:
5 3 8 6

FCFS Scheduling
PID AT  BT  CT  TAT WT
1   0   5   5   5   0
2   1   3   8   7   4
3   2   8  16  14   6
4   3   6  22  19  13

Average Turnaround Time = 11.25 ms
Average Waiting Time    = 5.75 ms
```

=>SJF(Non-preemptive):

```
#include <stdio.h>

struct Process
{
    int pid;
    int at;
    int bt;
    int ct;
    int tat;
    int wt;
    int done;
};

int main()
{
    int n;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++)
    {
        printf("\nEnter Arrival Time and Burst Time for P%d: ", i + 1);

        p[i].pid = i + 1;
        scanf("%d %d", &p[i].at, &p[i].bt);

        p[i].done = 0;
    }
}
```

```

int completed = 0;
int time = 0;

float totalWT = 0;
float totalTAT = 0;

while(completed < n)
{
    int idx = -1;
    int minBT = 9999;

    for(int i = 0; i < n; i++)
    {
        if(p[i].at <= time && p[i].done == 0)
        {
            if(p[i].bt < minBT)
            {
                minBT = p[i].bt;
                idx = i;
            }
        }
    }

    if(idx == -1)
    {
        time++;
    }
    else
    {
        time += p[idx].bt;

        p[idx].ct = time;
        p[idx].tat = p[idx].ct - p[idx].at;
        p[idx].wt = p[idx].tat - p[idx].bt;

        totalWT += p[idx].wt;
        totalTAT += p[idx].tat;

        p[idx].done = 1;
        completed++;
    }
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

```



```

for(int i = 0; i < n; i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid,
        p[i].at,
        p[i].bt,
        p[i].ct,
        p[i].tat,
        p[i].wt);
}

printf("\nAverage Turnaround Time = %.2f", totalTAT / n);
printf("\nAverage Waiting Time = %.2f\n", totalWT / n);

return 0;
}

```

```

Enter number of processes: 4

Enter Arrival Time and Burst Time for P1: 1 2

Enter Arrival Time and Burst Time for P2: 3 2

Enter Arrival Time and Burst Time for P3: 4 4

Enter Arrival Time and Burst Time for P4: 2 3

PID AT  BT  CT  TAT WT
P1  1   2   3   2   0
P2  3   2   5   2   0
P3  4   4  12   8   4
P4  2   3   8   6   3

Average Turnaround Time = 4.50
Average Waiting Time = 1.75

```

SJF(PREEMPTIVE):

```
#include <stdio.h>
```

```

struct Process {
int pid;
    int at;    // Arrival Time
int bt;    // Burst Time    int rt;
// Remaining Time    int ct;
// Completion Time    int tat;
// Turnaround Time    int wt;
// Waiting Time
};

```

```

int main() {
    int n;
    printf("1BF24CS337\n");

    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        printf("\nEnter Arrival Time and Burst Time for P%d: ", i + 1);

        p[i].pid = i + 1;
        scanf("%d %d", &p[i].at, &p[i].bt);

        p[i].rt = p[i].bt;
    }

    int completed = 0, time = 0;
    float totalWT = 0, totalTAT = 0;

    while(completed < n) {
        int idx = -1;    int
        minRT = 9999;

        // Find process with shortest remaining time
        for(int i = 0; i < n; i++) {
            if(p[i].at <=
            time && p[i].rt > 0) {
                if(p[i].rt <
                minRT) {
                    minRT = p[i].rt;
                    idx = i;
                }
            }
        }
    }
}

```

```

        // If no process has arrived
if(idx == -1) {      time++;
}      else {
        // Execute process for 1 unit time
p[idx].rt--;      time++;

        // If process completed
if(p[idx].rt == 0) {
completed++;

        p[idx].ct = time;
p[idx].tat = p[idx].ct - p[idx].at;
p[idx].wt = p[idx].tat - p[idx].bt;

        totalWT += p[idx].wt;
totalTAT += p[idx].tat;
    }
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

for(int i = 0; i < n; i++) {
printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid,
p[i].at,
p[i].bt,
p[i].ct,
p[i].tat,
p[i].wt);
}

```

```
printf("\nAverage Turnaround Time = %.2f", totalTAT / n);  
printf("\nAverage Waiting Time = %.2f\n", totalWT / n);
```

1BF24CS337

Enter number of processes: 4

Enter Arrival Time and Burst Time for P1: 1 2

Enter Arrival Time and Burst Time for P2: 2 4

Enter Arrival Time and Burst Time for P3: 1 3

Enter Arrival Time and Burst Time for P4: 2 5

PID	AT	BT	CT	TAT	WT
P1	1	2	3	2	0
P2	2	4	10	8	4
P3	1	3	6	5	2
P4	2	5	15	13	8

Average Turnaround Time = 7.00

Average Waiting Time = 3.50

PRIORITY(NON-PREEMPTIVE):

```
#include <stdio.h>
```

```
int main() {
```

```

int n, i, j;

printf("Enter number of processes: ");
scanf("%d", &n);

int pid[n], at[n], bt[n], pr[n];
int ct[n], tat[n], wt[n];   int
completed[n];

for(i = 0; i < n; i++)
{
    printf("\nEnter details for Process P%d\n", i + 1);

    pid[i] = i + 1;

    printf("Arrival Time: ");
    scanf("%d", &at[i]);

    printf("Burst Time: ");
    scanf("%d", &bt[i]);

    printf("Priority: ");
    scanf("%d", &pr[i]);

    completed[i] = 0;
}

int current_time = 0;
int completed_count = 0;

while(completed_count < n)
{

```

```

    int highest_priority = 9999;
int selected = -1;

    for(i = 0; i < n; i++)
    {
        if(at[i] <= current_time && completed[i] == 0)
        {
            if(pr[i] < highest_priority)
            {
                highest_priority = pr[i];
selected = i;
            }
        }
    }

    if(selected == -1)
    {
        current_time++;
    }
else
    {
        ct[selected] = current_time + bt[selected];

        tat[selected] = ct[selected] - at[selected];
wt[selected] = tat[selected] - bt[selected];

        current_time = ct[selected];

        completed[selected] = 1;
completed_count++;
    }
}

```

```

printf("\n----- Priority Scheduling (Non-Preemptive) ----- \n");

printf("\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\n");

float total_tat = 0, total_wt = 0;

for(i = 0; i < n; i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
           pid[i], at[i], bt[i], pr[i],
ct[i], tat[i], wt[i]);

    total_tat += tat[i];
total_wt += wt[i];
}

printf("\nAverage Turnaround Time = %.2f", total_tat / n);
printf("\nAverage Waiting Time = %.2f\n", total_wt / n);

return 0;
}

```



```

1BF24CS337
Enter number of processes: 4

Enter details for Process P1
Arrival Time: 1
Burst Time: 2
Priority: 2

Enter details for Process P2
Arrival Time: 1
Burst Time: 3
Priority: 1

Enter details for Process P3
Arrival Time: 2
Burst Time: 4
Priority: 3

Enter details for Process P4
Arrival Time: 2
Burst Time: 4
Priority: 4

----- Priority Scheduling (Non-Preemptive) -----
PID  AT   BT   PR   CT   TAT   WT
P1   1    2    2    6    5    3
P2   1    3    1    4    3    0
P3   2    4    3   10    8    4
P4   2    4    4   14   12    8

Average Turnaround Time = 7.00
Average Waiting Time = 3.75

```

PRIORITY(PREEMPTIVE):

```
#include <stdio.h>
```

```

int main() {
    int n, i;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int pid[n], at[n], bt[n], pr[n];
    int rt[n], ct[n], tat[n], wt[n];

```

```

for(i = 0; i < n; i++)
{
    printf("\nEnter details for Process P%d\n", i + 1);

    pid[i] = i + 1;

    printf("Arrival Time: ");
scanf("%d", &at[i]);

    printf("Burst Time: ");
scanf("%d", &bt[i]);

    printf("Priority: ");
scanf("%d", &pr[i]);
    rt[i] =
bt[i];
}

int current_time = 0;
int completed = 0;

while(completed < n)
{
    int highest_priority = 9999;
int selected = -1;

    for(i = 0; i < n; i++)
    {
        if(at[i] <= current_time && rt[i] > 0)
        {
            if(pr[i] < highest_priority)
            {

```

```

        highest_priority = pr[i];
selected = i;
    }
}
}

if(selected == -1)
{
    current_time++;
} else
{
rt[selected]--;

    current_time++;

    if(rt[selected] == 0)
    {
        ct[selected] = current_time;

        tat[selected] = ct[selected] - at[selected];

        wt[selected] = tat[selected] - bt[selected];

        completed++;
    }
}

printf("\n----- Priority Scheduling (Preemptive) ----- \n");
printf("\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\n");

float total_tat = 0, total_wt = 0;

```

```

    for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            pid[i], at[i], bt[i], pr[i],
            ct[i], tat[i], wt[i]);

        total_tat += tat[i];
        total_wt += wt[i];
    }

    printf("\nAverage Turnaround Time = %.2f", total_tat / n);
    printf("\nAverage Waiting Time = %.2f\n", total_wt / n);

    return 0;
}

```

```

Enter number of processes: 3

Enter details for Process P1
Arrival Time: 1
Burst Time: 2
Priority: 2

Enter details for Process P2
Arrival Time: 3
Burst Time: 2
Priority: 1

Enter details for Process P3
Arrival Time: 4
Burst Time: 2
Priority: 3

----- Priority Scheduling (Preemptive) -----

PID AT  BT  PR  CT  TAT WT
P1  1   2   2   3   2  0
P2  3   2   1   5   2  0
P3  4   2   3   7   3  1

Average Turnaround Time = 2.33
Average Waiting Time = 0.33

```

ROUND ROBIN:

```
#include <stdio.h>
```

```

int main() {
    int n, tq, i;
    printf("Enter number of processes: ");

```

```

scanf("%d", &n);

printf("Enter Time Quantum: ");
scanf("%d", &tq);

int pid[n], at[n], bt[n], rt[n];
int ct[n], tat[n], wt[n];

for(i = 0; i < n; i++)
{
    printf("\nEnter details for Process P%d\n", i + 1);

    pid[i] = i + 1;

    printf("Arrival Time: ");
    scanf("%d", &at[i]);

    printf("Burst Time: ");
    scanf("%d", &bt[i]);
    rt[i] =
bt[i];
}

int completed = 0, current_time = 0;

while(completed < n)
{
    int executed
= 0;

    for(i = 0; i < n; i++)
    {
        if(at[i] <= current_time && rt[i] > 0)

```

```

        {
executed = 1;
        if(rt[i] >
tq)
        {
current_time += tq;
rt[i] -= tq;        }
else
        {
current_time += rt[i];

        ct[i] = current_time;

        tat[i] = ct[i] - at[i];

        wt[i] = tat[i] - bt[i];
        rt[i] =
0;

        completed++;
        }
    }
}

if(executed == 0)
{
    current_time++;
}
}

printf("\n----- Round Robin Scheduling-----\n");

```

```

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

float total_tat = 0, total_wt = 0;

for(i = 0; i < n; i++)
{
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
           pid[i], at[i], bt[i],
           ct[i], tat[i], wt[i]);

    total_tat += tat[i];
    total_wt += wt[i];
}

printf("\nAverage Turnaround Time = %.2f",
       total_tat / n);

printf("\nAverage Waiting Time = %.2f\n",
       total_wt / n);

return 0;
}

```



```

1BF24CS337
Enter number of processes: 3
Enter Time Quantum: 2

Enter details for Process P1
Arrival Time: 1
Burst Time: 2

Enter details for Process P2
Arrival Time: 1
Burst Time: 4

Enter details for Process P3
Arrival Time: 2
Burst Time: 5

----- Round Robin Scheduling -----

PID  AT  BT  CT  TAT  WT
P1   1   2   3   2   0
P2   1   4   9   8   4
P3   2   5  12  10   5

Average Turnaround Time = 6.67
Average Waiting Time = 3.00

```

PROGRAM-2

MULTI-LEVEL QUEUE

```
#include <stdio.h>
```

```

struct Process {
int pid;
    int qn;    // Queue Number
int at;    // Arrival Time

```

```

    int bt;    // Burst Time
    int rt;    // Remaining Time
    int ct;    // Completion Time
    int tat;   // Turnaround Time
    int wt;    // Waiting Time
};

int main() {
    int n, tq;

    printf("1BF24CS337\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    printf("Enter Time Quantum for Queue 1: ");
    scanf("%d", &tq);

    // Input
    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;

        printf("\nEnter Queue No, Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d %d", &p[i].qn, &p[i].at, &p[i].bt);

        p[i].rt = p[i].bt;
    }

    int completed = 0, time = 0;
    float avgTAT = 0, avgWT = 0; //

```

```

while(completed < n) {

    int executed = 0;

    // Queue 1 - Round Robin (Higher Priority)
    for(int i = 0; i < n; i++) {

        if(p[i].qn == 1 && p[i].at <= time && p[i].rt > 0) {

            executed = 1;

            if(p[i].rt > tq) {
time += tq;
p[i].rt -= tq;
            } else
            {
                time +=
p[i].rt;          p[i].rt =
0;

                p[i].ct = time;
p[i].tat = p[i].ct - p[i].at;
p[i].wt = p[i].tat - p[i].bt;

                avgTAT += p[i].tat;
avgWT += p[i].wt;
completed++;
            }
        }
    }
}

```

```

// Queue 2 - FCFS (Executed only if Queue 1 empty)
if(!executed) {      int idx = -1;      int earliest =
9999;

    for(int i = 0; i < n; i++) {
        if(p[i].qn == 2 && p[i].at <= time && p[i].rt > 0) {
if(p[i].at < earliest) {      earliest = p[i].at;
idx = i;
        }
    }
}

    if(idx != -1) {
time += p[idx].rt;
p[idx].rt = 0;

        p[idx].ct = time;
p[idx].tat = p[idx].ct - p[idx].at;
p[idx].wt = p[idx].tat - p[idx].bt;

        avgTAT += p[idx].tat;
avgWT += p[idx].wt;
completed++;
    }
else {
time++;
    }
}

}

printf("\nPID\tQNo\tAT\tBT\tCT\tTAT\tWT\n");

```

```

    for(int i = 0; i < n; i++) {
printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
      p[i].pid,
      p[i].qn,
      p[i].at,
      p[i].bt,
      p[i].ct,
      p[i].tat,
      p[i].wt);
    }

    printf("\nAverage Turnaround Time = %.2f", avgTAT/n);
printf("\nAverage Waiting Time = %.2f\n", avgWT/n);

    return 0;

```

1BF24CS337

Enter number of processes: 4

Enter Time Quantum for Queue 1: 2

Enter Queue No, Arrival Time and Burst Time for P1: 1 4 2

Enter Queue No, Arrival Time and Burst Time for P2: 2 1 3

Enter Queue No, Arrival Time and Burst Time for P3: 1 3 4

Enter Queue No, Arrival Time and Burst Time for P4: 2 2 3

PID	QNo	AT	BT	CT	TAT	WT
P1	1	4	2	6	2	0
P2	2	1	3	4	3	0
P3	1	3	4	10	7	3
P4	2	2	3	13	11	8

Average Turnaround Time = 5.75

Average Waiting Time = 2.75

PROGRAM-3

RATE MONOTONIC

```
#include <stdio.h>

struct Process {
    int pid;    int
    burst;    int
    period;    int
    remaining;
};

int main() {
    int n, time, hyper = 20;

    printf("1BF24CS337\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        printf("\nEnter Burst Time and Period for P%d: ", i + 1);

        p[i].pid = i + 1;
        scanf("%d %d", &p[i].burst, &p[i].period);

        p[i].remaining = 0;
    }

    printf("\nGantt Chart:\n");

    for(time = 0; time < hyper; time++) {
```

```

        // Release process at start of period
for(int i = 0; i < n; i++) {
    if(time % p[i].period == 0) {
        p[i].remaining = p[i].burst;
    }
}

    int idx = -1;    int
minPeriod = 9999;

    // Higher priority = smaller period
for(int i = 0; i < n; i++) {
    if(p[i].remaining > 0 && p[i].period < minPeriod) {
        minPeriod = p[i].period;
        idx = i;
    }
}

    if(idx != -1) {
        printf("P%d ", p[idx].pid);
        p[idx].remaining--;
    }
    else {
        printf("Idle ");
    }
}

    return 0;
}

```


1BF24CS337

Enter number of processes: 4

Enter Burst Time and Period for P1: 1 2

Enter Burst Time and Period for P2: 2 4

Enter Burst Time and Period for P3: 2 3

Enter Burst Time and Period for P4: 1 5

Gantt Chart:

P1 P3 P1 P3 P1 P3 P1 P3 P1 P3 P1 P3 P1 P3 P1 P3 P3

Earliest-deadline First :

```
#include <stdio.h>
```

```
struct Process  
{   int pid;  
  int burst;   int  
  period;   int  
  deadline;   int  
  remaining;  
};
```

```
int main() {   int n,  
time, hyper = 20;  
printf("1BF24CS337\n");  
  printf("Enter number of processes: ");  
  scanf("%d", &n);  
  struct Process p[n];  
  
  for(int i = 0; i < n; i++) {      printf("\nEnter Burst Time
```

```

    and Period for P%d: ", i + 1);

    p[i].pid = i + 1;    scanf("%d %d",
&p[i].burst, &p[i].period);

    p[i].remaining = 0;
    p[i].deadline = p[i].period;}

printf("\nGantt Chart:\n");

for(time = 0; time < hyper; time++) {

    // Release tasks    for(int i = 0; i <
n; i++) {        if(time % p[i].period ==
0) {            p[i].remaining = p[i].burst;
p[i].deadline = time + p[i].period;
        }
    }

    int idx = -1;
    int earliest = 9999;

    // Select earliest deadline    for(int i = 0; i < n;
i++) {        if(p[i].remaining > 0 && p[i].deadline <
earliest) {            earliest = p[i].deadline;            idx
= i;
        }
    }

    if(idx != -1) {
printf("P%d ", p[idx].pid);
p[idx].remaining--;
    } else {
printf("Idle ");
    }
}

return 0;
}

```

1BF24CS337

Enter number of processes: 3

Enter Burst Time and Period for P1: 1 2

Enter Burst Time and Period for P2: 2 4

Enter Burst Time and Period for P3: 2 6

Gantt Chart:

P1 P2 P1 P2 P1 P3 P1 P2 P1 P2 P1 P2 P1 P2 P1 P3 P1 P2

Proportional scheduling

```
#include <stdio.h>
```

```
struct Process {  
    int pid;    int  
    share;  
};
```

```
int main() {  
    int n, total = 0;  
    printf("1BF24CS337\n");  
  
    printf("Enter number of processes: ");  
    scanf("%d", &n);
```

```
    struct Process p[n];
```

```
    for(int i = 0; i < n; i++) {        printf("\nEnter  
CPU share for P%d: ", i + 1);
```

```
        p[i].pid = i + 1;  
        scanf("%d", &p[i].share);
```

```

        total += p[i].share;
    }

    printf("\nCPU Allocation:\n");

    for(int i = 0; i < n; i++) {

        float percent =
            ((float)p[i].share / total) * 100;

        printf("P%d -> %.2f%% CPU Time\n",
            p[i].pid,
percent);
    }

    return 0;
}

```

```
1BF24CS337
Enter number of processes: 3

Enter CPU share for P1: 24

Enter CPU share for P2: 33

Enter CPU share for P3: 44

CPU Allocation:
P1 -> 23.76% CPU Time
P2 -> 32.67% CPU Time
P3 -> 43.56% CPU Time
```

PROGRAM-4

PRODUCER-CONSUMER PROBLEM USING SEMAPHORES

```
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0; int
out = 0;

int empty = BUFFER_SIZE;
int full = 0;
```

```

void produce() {
    if (empty == 0) {
        printf("\nBuffer is full! Cannot produce.\n");
    }
    return;

    int item;
    printf("Enter item to produce: ");
    scanf("%d", &item);

    buffer[in] = item;
    printf("Produced: %d at index %d\n", item, in);

    in = (in + 1) % BUFFER_SIZE;
    empty--;    full++;
}

void consume() {
    if (full == 0) {
        printf("\nBuffer is empty! Cannot consume.\n");
    }
    return;

    int item = buffer[out];
    printf("Consumed: %d from index %d\n", item, out);

    out = (out + 1) % BUFFER_SIZE;
    full--;
    empty++;
}

```

```

int main() {    int choice;
printf("1BF24CS337\n");

    printf("Buffer Size: %d\n", BUFFER_SIZE);

    while (1) {
        printf("\n1. Produce\n2. Consume\n3. Exit\n");
        printf("Current State: [Full slots: %d | Empty slots: %d]\n", full, empty);
printf("Enter choice: ");

        if (scanf("%d", &choice) != 1) break;

        switch (choice) {
case 1:            produce();
break;            case 2:
consume();        break;
case 3:
printf("Exiting...\n");
exit(0);        default:
                printf("Invalid choice!\n");
        }
    }

    return 0;
}

```

1BF24CS337

Buffer Size: 5

1. Produce

2. Consume

3. Exit

Current State: [Full slots: 0 | Empty slots: 5]

Enter choice: 1

Enter item to produce: 2

Produced: 2 at index 0

1. Produce

2. Consume

3. Exit

Current State: [Full slots: 1 | Empty slots: 4]

Enter choice: 1

Enter item to produce: 3

Produced: 3 at index 1

1. Produce

2. Consume

3. Exit

Current State: [Full slots: 2 | Empty slots: 3]

Enter choice: 2

Consumed: 2 from index 0

DINNING PHILOSOPHERS PROBLEM

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
#define N 5 // Number of philosophers
pthread_mutex_t forks[N]; // One mutex per
fork pthread_t philosophers[N]; void*
philosopher(void* num)
{
int id = *(int*)num; int left
= id; // left fork index
int right = (id + 1) % N; // right fork index
while (1) {
```

```

printf("Philosopher %d is thinking.\n", id);
sleep(1); // Thinking
// To avoid deadlock, philosophers with even id pick left then right, //
odd id pick right then left.
if (id % 2 == 0)
{
// Pick up left fork
pthread_mutex_lock(&forks[left]);
printf("Philosopher %d picked up left fork %d.\n", id, left);
// Pick up right fork

pthread_mutex_lock(&forks[right]);
printf("Philosopher %d picked up right fork %d.\n", id, right); } else
{ // Pick up right fork
pthread_mutex_lock(&forks[right]);
printf("Philosopher %d picked up right fork %d.\n", id, right); // Pick up left fork
pthread_mutex_lock(&forks[left]);
printf("Philosopher %d picked up left fork %d.\n", id, left);
} //
Eating
printf("Philosopher %d is eating.\n", id); sleep(2);
// Put down forks
pthread_mutex_unlock(&forks[left]);
pthread_mutex_unlock(&forks[right]);
printf("Philosopher %d put down forks %d and %d.\n", id, left, right);
} return
NULL;
}

int main() {
int i;

```

```

int ids[N]; // Initialize forks (mutexes) for
(i = 0; i < N; i++)
{ pthread_mutex_init(&forks[i], NULL);
  ids[i] = i;
}
// Create philosopher threads for
(i = 0; i < N; i++)

{
pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
} // Join threads (will never happen here, but good practice) for
(i = 0; i < N; i++)
{
pthread_join(philosophers[i], NULL); }
// Destroy mutexes (unreachable here)
for (i = 0; i < N; i++)
{
pthread_mutex_destroy(&forks[i]);
} return
0;
}

```

```
Produced: 0 at buffer[0]
Consumed: 0 from buffer[0]
Produced: 1 at buffer[1]
Produced: 2 at buffer[2]
Consumed: 1 from buffer[1]
Produced: 3 at buffer[3]
Consumed: 2 from buffer[2]
Produced: 4 at buffer[4]
Produced: 5 at buffer[0]
Consumed: 3 from buffer[3]
Produced: 6 at buffer[1]
Produced: 7 at buffer[2]
Consumed: 4 from buffer[4]
Produced: 8 at buffer[3]
Produced: 9 at buffer[4]
Consumed: 5 from buffer[0]
Produced: 10 at buffer[0]
Consumed: 6 from buffer[1]
Produced: 11 at buffer[1]
Consumed: 7 from buffer[2]
Produced: 12 at buffer[2]
Consumed: 8 from buffer[3]
Produced: 13 at buffer[3]
Consumed: 9 from buffer[4]
Produced: 14 at buffer[4]
Consumed: 10 from buffer[0]
Produced: 15 at buffer[0]
Consumed: 11 from buffer[1]
Produced: 16 at buffer[1]
Consumed: 12 from buffer[2]
Produced: 17 at buffer[2]
Consumed: 13 from buffer[3]
Produced: 18 at buffer[3]
```

PROGRAM-5 BANKER'S ALGORITHM

```
#include<stdio.h>

int main() {
    int allocation[10][10], max[10][10], need[10][10];
    int available[10], finish[10], safeSequence[10];
    int i, j, k, p, r, count = 0;

    printf("1BF24CS337\n");

    printf("Enter number of processes: ");
    scanf("%d", &p);

    printf("Enter number of resources: ");
    scanf("%d", &r);

    printf("Enter Allocation Matrix:\n");
    for(i = 0; i < p; i++)
    {
        for(j = 0; j < r;
j++)
        {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("Enter Maximum Demand Matrix:\n");
    for(i = 0; i < p; i++)
    {
        for(j = 0; j < r;
j++)
        {
            scanf("%d", &max[i][j]);
        }
    }
```

```

    printf("Enter Available Resources:\n");
for(i = 0; i < r; i++)
    {
        scanf("%d", &available[i]);
    }

    // Calculate Need Matrix
for(i = 0; i < p; i++)
    {
        for(j = 0; j < r;
j++)
            {
                need[i][j] = max[i][j] - allocation[i][j];
            }
    }

    // Initialize finish array
for(i = 0; i < p; i++)
    {
        finish[i]
= 0;
    }

    while(count < p)
    {
        int found
= 0;

        for(i = 0; i < p; i++)
            {
                if(finish[i] == 0)
                    {
                        for(j = 0;
j < r; j++)
                            {

```

```

        if(need[i][j] > available[j])
        {
break;
        }
    }
    if(j == r)
    {
        for(k = 0; k < r; k++)
        {
            available[k] += allocation[i][k];
        }

        safeSequence[count] = i;
count++;        finish[i] = 1;
found = 1;
    }
}
}

    if(found == 0)
    {
        printf("\nSystem is not in a safe state.\n");
return 0;
    }
}

    printf("\nSystem is in a safe state.\n");
printf("Safe sequence is: ");

    for(i = 0; i < p; i++)
    {
        printf("P%d", safeSequence[i]);

```

```

    if(i != p - 1)
    {
        printf(" -> ");
    }
}

return 0;
}

```

```

1BF24CS337
Enter number of processes: 5
Enter number of resources: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2
Process returned 0 (0x0)    execution time : 58.771 s

```


DEADLOCK DETECTION:

```
#include<stdio.h>
```

```
int main() {
    int allocation[10][10], request[10][10];
    int available[10];    int finish[10],
    safeSequence[10];    int i, j, k, p, r, count
    = 0;

    printf("USN-1BF24CS337\n");
    printf("Enter the number of processes: ");
    scanf("%d", &p);

    printf("Enter the number of resources: ");
    scanf("%d", &r);

    // Allocation Matrix    printf("Enter
the allocation matrix:\n");    for(i = 0; i <
p; i++)
    {        printf("Process %d:
", i);

        for(j = 0; j < r; j++)
        {
            scanf("%d", &allocation[i][j]);
        }
    }

    // Request Matrix    printf("Enter
the request matrix:\n");    for(i = 0; i <
p; i++)
    {
        printf("Process %d: ", i);
        for(j = 0; j < r; j++)
        {
            scanf("%d", &request[i][j]);
        }
    }
}
```

```

// Available Resources    printf("Enter
the available resources: ");    for(i = 0; i <
r; i++)
{
    scanf("%d", &available[i]);
}

// Initialize finish array
for(i = 0; i < p; i++)
{
    finish[i]
= 0;
}

while(count < p)
{
    int found
= 0;

    for(i = 0; i < p; i++)
    {
        if(finish[i] == 0)
        {
            for(j = 0;
j < r; j++)
            {
                if(request[i][j] > available[j])
                {
                    break;
                }
            }
            if(j == r)
            {
                for(k = 0; k < r;
k++)
                {
                    available[k] += allocation[i][k];

```

```

    }

    safeSequence[count++] = i;
finish[i] = 1;        found = 1;
    }
    }
}

if(found == 0)
{
break;
}
}

if(count == p)
{
    printf("\nSystem is in safe state.\n");
printf("Safe Sequence is: ");

    for(i = 0; i < p; i++)
    {
        printf("P%d ", safeSequence[i]);
    }
}
else
{
    printf("\nDeadlock detected in the system.\n");
}

return 0;
}

```

```
"C:\Users\Yogesh Kumar Shec  X + v
1BF24CS337
Enter number of processes: 4
Enter number of resource types: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 1 2
2 1 1
Enter Maximum Matrix:
2 1 3
0 1 2
0 4 5
1 3 0
Enter Available Resources:
3 2 2

System is in SAFE state.
Safe Sequence: P1 P3 P0 P2

Process returned 0 (0x0)   execution time : 50.010 s
Press any key to continue.
```

PROGRAM-6

FIRST FIT, BEST FIT, WORST FIT

```
#include <stdio.h>
```

```
void firstFit(int blockSize[], int blocks, int processSize[], int processes) {  
    int allocation[processes];
```

```
        for (int i = 0; i < processes; i++)  
            allocation[i] = -1;
```

```
        int used[blocks];    for (int i  
        = 0; i < blocks; i++)  
            used[i] = 0;
```

```

        for (int i = 0; i < processes; i++) {
for (int j = 0; j < blocks; j++) {

        // block must be unused
        if (!used[j] && blockSize[j] >= processSize[i]) {
allocation[i] = j;
        used[j] = 1; // mark block as used
break;
        }
    }
}

printf("\n--- First Fit ---\n");
printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < processes; i++) {
printf("%d\t%d\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

void bestFit(int blockSize[], int blocks, int processSize[], int processes) {
int allocation[processes];

    for (int i = 0; i < processes; i++)
allocation[i] = -1;

```

```

    int used[blocks];    for (int i
= 0; i < blocks; i++)
used[i] = 0;

    for (int i = 0; i < processes; i++) {
int bestIdx = -1;

    for (int j = 0; j < blocks; j++) {

        if (!used[j] && blockSize[j] >= processSize[i]) {

            if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx]) {
bestIdx = j;
            }
        }
    }

    if (bestIdx != -1) {
allocation[i] = bestIdx;
used[bestIdx] = 1;
    }
}

printf("\n--- Best Fit ---\n");
printf("Process No.\tProcess Size\tBlock No.\n");

    for (int i = 0; i < processes; i++) {
printf("%d\t\t%d\t\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
printf("%d\n", allocation[i] + 1);
    else

```



```

        printf("Not Allocated\n");
    }
}

void worstFit(int blockSize[], int blocks, int processSize[], int processes) {
    int allocation[processes];

    for (int i = 0; i < processes; i++)
        allocation[i] = -1;

    int used[blocks];    for (int i
= 0; i < blocks; i++)
        used[i] = 0;

    for (int i = 0; i < processes; i++) {
        int worstIdx = -1;

        for (int j = 0; j < blocks; j++) {

            if (!used[j] && blockSize[j] >= processSize[i]) {

                if (worstIdx == -1 || blockSize[j] > blockSize[worstIdx]) {
                    worstIdx = j;
                }
            }
        }

        if (worstIdx != -1) {
            allocation[i] = worstIdx;
            used[worstIdx] = 1;
        }
    }
}

```

```

printf("\n--- Worst Fit ---\n");
printf("Process No.\tProcess Size\tBlock No.\n");

for (int i = 0; i < processes; i++) {
printf("%d\t\t%d\t\t", i + 1, processSize[i]);

    if (allocation[i] != -1)
printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
    }
}

int main() {    int blocks,
processes;
printf("IBF24CS337\n");

    printf("Enter number of memory blocks: ");
scanf("%d", &blocks);

    int blockSize[blocks];

    printf("Enter sizes of memory blocks:\n");
for (int i = 0; i < blocks; i++) {
scanf("%d", &blockSize[i]);
    }

    printf("Enter number of processes: ");
scanf("%d", &processes);

    int processSize[processes];

```

```
    printf("Enter sizes of processes:\n");
    for (int i = 0; i < processes; i++) {
        scanf("%d", &processSize[i]);
    }

    firstFit(blockSize, blocks, processSize, processes);
    bestFit(blockSize, blocks, processSize, processes);
    worstFit(blockSize, blocks, processSize, processes);

    return 0;
}
```

```
"C:\Users\Yogesh Kumar Shec  × + v
1BF24CS337
Enter number of memory blocks: 5
Enter sizes of memory blocks:
100 500 200 300 600
Enter number of processes: 4
Enter sizes of processes:
212 417 112 426

--- First Fit ---
Process No.      Process Size      Block No.
1                212                2
2                417                5
3                112                3
4                426                Not Allocated

--- Best Fit ---
Process No.      Process Size      Block No.
1                212                4
2                417                2
3                112                3
4                426                5

--- Worst Fit ---
Process No.      Process Size      Block No.
1                212                5
2                417                2
3                112                4
4                426                Not Allocated

Process returned 0 (0x0)   execution time : 43.563 s
Press any key to continue.
```

PROGRAM-7

Write a C program to simulate page replacement algorithms. a) FIFO b) LRU
c) Optimal

```
#include <stdio.h>
```

```
#define MAX 50
```

```

// Function to check if page exists in frames int
isPresent(int frames[], int num_frames, int page) {

for (int i = 0; i < num_frames; i++)
    {        if (frames[i] ==
page)        return 1;    }
return 0;
}

// Function to print frames void printFrames(int
frames[], int num_frames)
{    printf("[");    for (int i = 0; i <
num_frames; i++)
    {        if (frames[i] != -1)
printf("%d", frames[i]);
else        printf(" ");

        if (i != num_frames - 1)
printf(" ");    }
printf("]\n");
}
// ----- FIFO ----- void
FIFO(int pages[], int n, int num_frames)
{    int
frames[MAX];    int
page_faults = 0;
int next = 0;

    for (int i = 0; i < num_frames; i++)
frames[i] = -1;

    printf("\n--- FIFO Page Replacement ---\n\n");

    for (int i = 0; i < n; i++)

```

```

    {
        int page = pages[i];

        if (!isPresent(frames, num_frames, page))
        {
            frames[next] = page;        next
            = (next + 1) % num_frames;
            page_faults++;
        }

        printf("Page %d -> ", page);
        printFrames(frames, num_frames);
    }

    printf("\nTotal Page Faults (FIFO): %d\n", page_faults);
}

// ----- LRU ----- void
LRU(int pages[], int n, int num_frames)
{
    int frames[MAX];
    int last_used[MAX];
    int page_faults = 0;

    for (int i = 0; i < num_frames; i++)
    {
        frames[i] = -
1;        last_used[i] =
-1;
    }

    printf("\n--- LRU Page Replacement ---\n\n");

    for (int i = 0; i < n; i++)

```

```

    {        int page =
pages[i];        int found
= 0;

        // Check if page exists        for (int
j = 0; j < num_frames; j++)
    {        if (frames[j]
== page)
        {        found
= 1;
last_used[j] = i;
break;
        }
    }

        // Page Fault
if (!found)    {
int pos = -1;

        // Empty frame available
for (int j = 0; j < num_frames; j++)
    {        if
(frames[j] == -1)        {
pos = j;        break;
        }
    }

        // Replace least recently used
if (pos == -1)
    {
        int min = last_used[0];
pos = 0;

```

```

        for (int j = 1; j < num_frames; j++)
        {
            if (last_used[j] < min)
            {
                min = last_used[j];
pos = j;
            }
        }

        frames[pos] = page;
last_used[pos] = i;
page_faults++;
    }

    printf("Page %d -> ", page);
printFrames(frames, num_frames);
}

printf("\nTotal Page Faults (LRU): %d\n", page_faults); }

// ----- Optimal ----- void
Optimal(int pages[], int n, int num_frames)
{
    int
frames[MAX];    int
page_faults = 0;

    for (int i = 0; i < num_frames; i++)
frames[i] = -1;

    printf("\n--- Optimal Page Replacement ---\n\n");

```



```

    for (int i = 0; i < n; i++)
    {
        int page =
pages[i];

        if (!isPresent(frames, num_frames, page))
        {
            int
pos = -1;

            // Empty frame
            for (int j =
0; j < num_frames; j++)
            {
                if
(frames[j] == -1)
                {
pos = j;
break;
                }
            }

            // Find optimal replacement
            if (pos == -1)
            {
                int farthest = -
1;

                for (int j = 0; j < num_frames; j++)
                {
                    int k;
for (k = i + 1; k < n; k++)
                {
                    if (frames[j] == pages[k])
break;
                }
            }

```

```

        // Page never used again
if (k == n)      {
pos = j;         break;
                }

        // Farthest use
if (k > farthest)
{
    farthest =
k;           pos = j;
            }
        }
    }

    frames[pos] = page;
page_faults++;
}

    printf("Page %d -> ", page);
printFrames(frames, num_frames);
}

    printf("\nTotal Page Faults (Optimal): %d\n", page_faults);
}

int main() {    int n, num_frames;
int pages[MAX];
printf("1BF24CS337\n");

    printf("Enter number of pages: ");
scanf("%d", &n);

```

```
    printf("Enter page reference string:\n");  
    for (int i = 0; i < n; i++)    scanf("%d",  
    &pages[i]);
```

```
    printf("Enter number of frames: ");  
    scanf("%d", &num_frames);
```

```
    FIFO(pages, n, num_frames);  
    LRU(pages, n, num_frames);  
    Optimal(pages, n, num_frames);
```

```
    return 0;  
}
```

```
"C:\Users\Yogesh Kumar Shec  X  +
1BF24CS337
Enter number of pages: 12
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3
Enter number of frames: 3

--- FIFO Page Replacement ---

Page 7 -> [7    ]
Page 0 -> [7 0  ]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 3 1]
Page 0 -> [2 3 0]
Page 4 -> [4 3 0]
Page 2 -> [4 2 0]
Page 3 -> [4 2 3]
Page 0 -> [0 2 3]
Page 3 -> [0 2 3]

Total Page Faults (FIFO): 10

--- LRU Page Replacement ---

Page 7 -> [7    ]
Page 0 -> [7 0  ]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [4 0 3]
Page 2 -> [4 0 2]
Page 3 -> [4 3 2]
Page 0 -> [0 3 2]
Page 3 -> [0 3 2]

Total Page Faults (LRU): 9
```

--- Optimal Page Replacement ---

Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [2 4 3]
Page 2 -> [2 4 3]
Page 3 -> [2 4 3]
Page 0 -> [0 4 3]
Page 3 -> [0 4 3]

Total Page Faults (Optimal): 7